

obstructing airflow. However, entry-level or low-tier cabinets aren't fun when it comes to cable management. You have to spend some time in routing them appropriately and tying them down with cable ties. Cutouts will be present on these cabinets so it shouldn't be difficult. Make sure that you're routing all the PSU cables behind the motherboard plate. You can take advantage of the cutouts to route the specific cables to the front of the motherboard. Essentially, you should avoid connecting the cables directly in the motherboard area. This way you prevent any cable from dangling in the cabinet, obstructing the air intake from the front. Investing a few extra minutes in the beginning can save you a lot of time later. Opting for a semi-modular or modular PSU is also recommended. You will end up using only the cables you actually need.

Some modular cabinets allow you to remove the drive cages. If you aren't using them, it would be best to remove them. For example, if you have HDD mounts behind the motherboard plate, you can use them instead of the drive cages to offer better airflow.

Optical drive or hard drive cages are sometimes riveted to the cabinet, which means they have to be cut out. If you do so, make sure to not damage your cabinet. Just make sure you're not going to change your mind and want an optical drive later. The difference in airflow is significant though, so consider it.

Cabinet manufacturers will generally ship one or more fans inside. One fan is

clearly not enough to keep your system cool. Even though there are more fans already installed, you might not achieve the required level of cooling. In that case, you have to opt for aftermarket fans. For this guide, we are looking at air cooling. Hence, our focus will be more towards high airflow fans. Unless, your cabinet has an HDD cage obstructing the fan mount. Then you have to go for a



Dust build up will make your components heat up quickly

high static pressure fan. To keep things easier, let's go with only high airflow fans.

Consider a system with three fans for intake and two fans acting as exhaust. Referring to the specifications of the fans, the total CFM of the intake and exhaust fans can be calculated. If you're using the same fan model, then you can simply count the number of fans to determine whether there's higher intake or exhaust. If the intake CFM is higher than exhaust, it will result in positive pressure inside the cabinet. This means that more air is being pulled in than the amount being pushed out. Theoretically,

you will have hot air spending more time inside raising the temperature gradually which is negligible. However, positive air pressure will ensure that lower amount of dust enters your cabinet. Moving on, if the intake CFM is lower than exhaust, we will have negative pressure. Here, more air is leaving the cabinet compared to the air coming inside. This could result into vacuum forming inside the

temperatures later. If your surrounding has a lot of dust floating around, it would be best to go for a higher CFM intake configuration.



## FINAL THOUGHTS

By now you must have learned about the basics of airflow inside cabinets. You understand how the intake and exhaust work together to keep your components cool. Cable management is something you shouldn't ignore, especially when you're dealing with entry-level cabinets. You also came across important specifications of fans that matter while building a PC. Taking things to the next level would be actually calculating the intake and exhaust CFM. In this way, you can accurately determine whether there's positive or negative air pressure inside the cabinet. Then you could move on to running benchmarks to measure CPU and GPU temperatures.

Accordingly, you can adjust the number of intake and exhaust fans. If you wish to exert more control, you can install PWM fans. These fans will enable you to control the fan speed so that you can fine tune the air pressure inside. For aesthetic purposes, you can opt for LED fans (RGB fans if you're that kind of a person) since the heat generated is negligible. The possibilities are endless.

If you consider aftermarket solutions, make sure that you're reading the specifications before making purchases. It's true that better cooling will result in better performance in your components. You will also give all your precious components a longer lease of life. 